Satellite orbit

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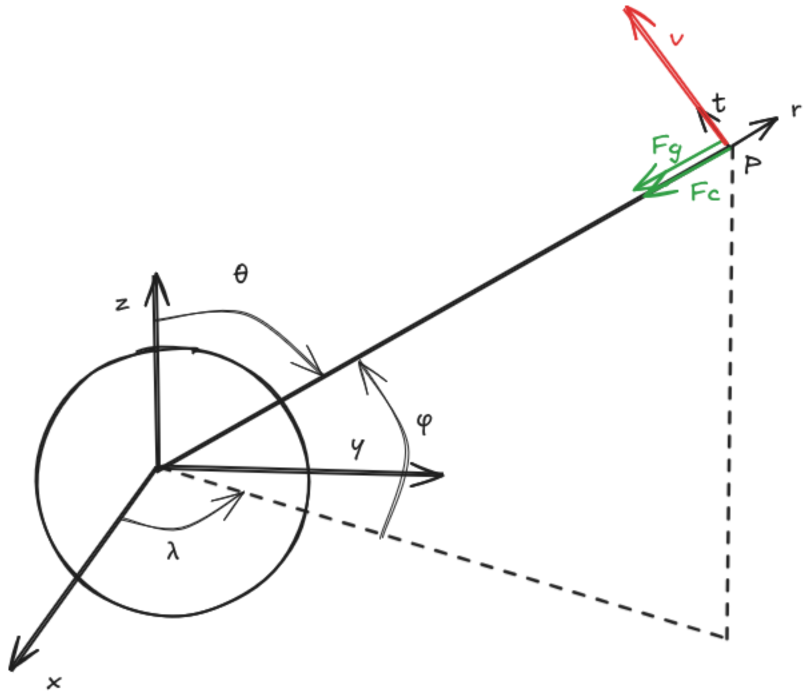
Contents

1 Introduction

Study how to maintain a satellite at a certain orbit

2 Schema

A satellite of a mass m is orbiting around a planet of mass M at altitude ρ [meters] from its center with a tangential speed of v m/s.



The satellite position in a spherical referential (ρ, θ, λ) is :

$$\vec{OS} = \begin{bmatrix} l \\ m \\ n \end{bmatrix}_{R1} = \rho \cdot \begin{bmatrix} \sin(\theta) \cdot \cos(\lambda) \\ \sin(\theta) \cdot \sin(\lambda) \\ \cos(\theta) \end{bmatrix}_{R1} \tag{1}$$

with

- ρ : radial distance from the center of the planet (origin) to the object, in meters.
- λ : longitude, angle of rotation on the z axis
- ϕ : latitude of the satellite, elevation angle from the plane formed by x and y axes. (elevation angle, altitude angle)
- θ : polar angle between the polar axis(here the z axis) and the radial distance to the satellite. Or angle between the zenith and the latitude of the satellite at altitude ϕ Complementary to ϕ. (inclination angle, zenith angle,) $\theta = \frac{\pi}{2} - \phi$
- λ : longitude, which is the azimuthal angle, which angle of rotation around the polar axis

These angles are chosen to match the angles used for GPS location on Earth : latitude ϕ and longitude λ
TODO : express the velocity vector of the satellite

3 Stable orbit

3.1 Problem

An artificial object (i.e a the satellite or a space craft)is orbiting around a space body at speed v and at altitude h_{sat}. Calculate the force needed to maintain the object at that specific altitude

3.2 Resolution

Without any force applied on the artificial object, it will continue moving in the same direction as the speed vector. To maintain it constant altitude, i.e. forcing a circular trajectory around the planet, a centripetal force is applied toward the space body (S) center :

$$F_c = \frac{m_{sat} \cdot v_t^2}{\rho} = \frac{m_{sat} \cdot v_t^2}{S_{radius} + h_{sat}} \tag{2}$$

$$F_c = m_{sat} \cdot \omega^2 \cdot \rho = m \cdot (\frac{2\pi}{T})^2 \cdot (S_{radius} + h_{sat}) \tag{3}$$

Todo : create table at the beginning describing each variables and its unit,
m_{sat} : mass of the orbiting object, in kg
ρ : distance in m from the center of the rotating object (planet, sun, moon) to the orbiting object
v_t : tangential velocity of the orbiting object, in m/s
S_{radii} : Space body radii, in m
h_{sat} : distance from the surface of the space body to the satellite, in m
$\omega = \frac{2\pi}{T}$: angular speed of the space body, in rad/s
T : orbital period, in s

In space, there are several ways to apply this force, like :

- gravity force, if the orbiting object is within the gravity field of the space body.
- thrusts applying force equal to F_c at given time

CASE A : Gravity force
A.1. compute spped to stay at a given altitude
A.2. compute geostationary altitude, for different planets (check cases where the space

body is gas and the satellite must orbit inside it)
CASE B : thrusters force :
compute fuel needed to maintain this (like for the ISS)

3.3 Case A : Satellite within space body's gravity field

$$F_c = m_{sat} \cdot g \tag{4}$$

$$m_{sat} \cdot \frac{v_t^2}{\rho} = m_{sat} \cdot \omega^2 \cdot \rho = m_{sat} \cdot g \tag{5}$$

$$\frac{v_t^2}{\rho} = \omega^2 \cdot \rho = g = \frac{G \cdot M_S}{\rho^2} \tag{6}$$

$$v_t = \sqrt{\frac{G \cdot M_S}{\rho}} \tag{7}$$

$$\omega = \sqrt{\frac{G \cdot M_S}{\rho^3}} \tag{8}$$

$$\rho = \frac{G \cdot M_S}{v_t^2} \tag{9}$$

$$\rho = \sqrt[3]{\frac{G \cdot M_S}{\omega^2}} \tag{10}$$

choose v_t (or ω) to stay at a specific orbit. Or if the orbiting object must have a specific speed, the altitude that it must reach is computed. Then, it is possible to check if the orbit at that altitude is free and can be used without colliding with other objects.

3.4 Geostationary orbit

Orbit where an object is moving at the same speed as the planet, and is always in the same position on the sky. if it is satellite used for observation, it will look at exactly the same location on Earth. Some satellite follow this orbit, mainly for international communication. to maintain a the orbiting object, the object has the same angular speed as the space body $\omega = \omega_S$

$$\omega = \omega_S \tag{11}$$

$$\rho_{go} = \sqrt[3]{\frac{G \cdot M_{SB}}{\omega_S^2}} \tag{12}$$

$$v_{go} = \omega_S \cdot r_{go} \tag{13}$$

Example for Earth : Period T of 23h, 56min and 4s

$$T = 24 * 3600 + 56 * 60 + 4 \tag{14}$$
$$\omega = \omega_{Earth} = \frac{2 \cdot \pi}{T} = \frac{2 \cdot \pi}{23h56min4s} \tag{15}$$
$$\rho_{go} = \sqrt[3]{\frac{G \cdot M_{SB}}{\omega^2}} \tag{16}$$
$$v_{go} = \omega_S \cdot r_{go} \tag{17}$$

As long as the orbiting object maintain this speed, it will stay at this orbit. However, friction with the high atmosphere might deviate the satellite from its path. some of them are equipped with thrusters to correct the trajectory when needed.
Application example : - telecommunication satellites, placed around the planet to ensure that it is fully covered - observation satellite : some satellites are placed to observe the same points of interest

3.5 Case B : Orbit outside gravity field

Altitude when gravity is very low

$$g = \frac{G \cdot M_S}{\rho^2} <= 0.001 \tag{18}$$
$$\rho > \sqrt{\frac{G \cdot M_S}{g}} \tag{19}$$

Satellite is attracted by other space bodies, so thrusters and or boosters are needed to maintain the expected path.

$$\vec{v} = \begin{bmatrix} v_\rho \\ v_\theta \\ v_\lambda \end{bmatrix}_{R1} \tag{20}$$

The components v_θ and v_λ depends on the v need to stay on orbit will maintain the satellite at orbit.

$$v = \sqrt{v_\theta^2 + v_\lambda^2} \tag{21}$$

The component v_ρ must be compensated by activating thrusters and boosters to keep the altitude ρ at the same value.

3.6 les constantes

- G :

- altitude de la position de départ
- k : coefficient caractéristique de la géométrie du solide
- C_x
- S : surface à l'air
-

$$\vec{B_1B_2}_{R_1} = \begin{bmatrix} v_1 \cdot \cos(\beta_4) \\ 0 \\ v_1 \cdot \sin(\beta_4) \end{bmatrix}_{R1} \quad \vec{B_1C_1}_{R_1} = \begin{bmatrix} c_1 \cdot cos(\beta_1) \\ 0 \\ c_1 \cdot sin(\beta_1) \end{bmatrix}_{R1} \quad \vec{C_1C}_{R_1} = \begin{bmatrix} c \cdot cos(\beta_1) \\ 0 \\ c \cdot sin(\beta_1) \end{bmatrix}_{R1} \tag{22}$$

$$\vec{C_1C_2} = \begin{bmatrix} v_2 \cdot cos(\gamma_1) \\ 0 \\ v_2 \cdot sin(\gamma_1) \end{bmatrix}_{R1} \quad \vec{CC_2}_{R_1} = \begin{bmatrix} -c_2 \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{C_2D}_{R_2} = \begin{bmatrix} -d \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{DD_1}_{R_3} = \begin{bmatrix} -d_1 \\ 0 \\ 0 \end{bmatrix}_{R_3} \tag{23}$$

$$\vec{D_1D_2}_{R_3} = \begin{bmatrix} -d_2 \cdot cos(\delta_1) \\ 0 \\ d_2 \cdot sin(\delta_1) \end{bmatrix}_{R_3} \tag{24}$$

-
-
-

L : longueur représentative de la fusée

$$ggagsfaaf \tag{25}$$

4 Formules utiles

nombre Reynolds

$$Re = \frac{\rho_{air} \cdot L \cdot v_{fusee}}{\mu} \tag{26}$$

accélération terrestre

$$g = G\frac{m_{terre}}{(rayon_terre + altitude_{z_0} + z)^2} \tag{27}$$

Force de pesanteur

$$Fg = m_fusee \cdot g \tag{28}$$

frottement laminaire

$$Ffl = -k * \eta * v_{fusee} \tag{29}$$

frottement turbulant

$$Fft = -\frac{1}{2} \cdot \rho_{air} \cdot C_x \cdot S \cdot v_{fusee}^2 \tag{30}$$

quantité de mouvement conservée

$$p_{fusee} = p_{gaz} \tag{31}$$

$$P_{fusee} = m_{fusee} * v_{fusee} \tag{32}$$

$$P_{gaz} = m_{gaz} * v_{gaz} \tag{33}$$

$$dp/dt = sommeF = m_{fusee} * a_{fusse} + dm_{fusee}/dt + v_{fusee} \tag{34}$$

$$dp/dt = m_gaz * a_gaz + dm_gaz/dt * v_{gaz} = F_{poussee} \tag{35}$$

equilibre des forces

$$\sum F = F_{poussée} - F_g - F_{frottement} \tag{36}$$

$$a = \frac{\sum F}{m_{fusee}} \tag{37}$$

Les formules standards de la cinématique donnent vitesse et position.

$$v_{fusee} = a \cdot \Delta t \tag{38}$$

$$z = a \cdot \Delta t^2 + v_{fusee} \cdot \Delta t + z_0 \tag{39}$$

5 Procédure de calcul

itérer à chaque pas de temps
calculer Re selon valeur Re, le frottement est laminaire ou turbulant. limite environ vers 3000
force gravité les frottements équilibre des forces accélération pas de temps vitesse positition

6 Calcul plus poussé

masse variable quantité de mouvement et poussée des gaz. perte de masse à cause des gaz
g change selon l'altitude
les propriétés de l'air change, ρ, μ, η, ...
L de la fusée si elle perd des étages
calcul précis du C_x : cf TAero, ou cours mécanique fluide incompressible EPFL

7 Conclusion

8 Other cases

1. establish the equation of the general movement of a satellite (the same as on Earth)
Acceleration

$$\vec{a} = \frac{\sum \vec{F}}{m_{sat}} \tag{40}$$

Force equilibrium in a spherical referential (check if this is correct. maybe another force must be added. Centripetal force is only tangential)

$$\sum \vec{F} = \vec{F_c} - \vec{F_g} = \vec{F_R} = \begin{bmatrix} F_c \cdot cos(\alpha) \\ F_c \cdot sin(\alpha) \\ 0 \end{bmatrix}_{R1} + \begin{bmatrix} -F_g \\ 0 \\ 0 \end{bmatrix}_{R1} = m_{sat} \cdot \begin{bmatrix} a_r \\ a_t \\ a_z \end{bmatrix}_{R1} \tag{41}$$

$$F_c \cdot cos(\alpha) = m_{sat} \cdot g \tag{42}$$

$$m_{sat} \cdot \frac{v_t^2}{r} \cdot cos(\alpha) = m_{sat} \cdot \omega^2 * r \cdot cos(\alpha) = m_{sat} \cdot g \tag{43}$$

$$\frac{v_t^2}{r} \cdot cos(\alpha) = \omega^2 \cdot r \cdot cos(\alpha) = g = \frac{G \cdot m_{Earth}}{r^2} \tag{44}$$

$$v_t = \sqrt{\frac{G \cdot m_{Earth}}{r \cdot cos(\alpha)}} \tag{45}$$

$$\omega = \sqrt{\frac{G \cdot m_{Earth}}{r^3 \cdot cos(\alpha)}} \tag{46}$$

$$\tag{47}$$

2. Drag and lift forces depend if there is an atmosphere or not at that altitude -> compute air density depending on altitude. it will become 0 in space 3. simulate different cases with different propulsion, ...

4. test case : propulsion force that must be applied to maintain orbit over the planet (Earth)
5. the same case but using gravity force
6. so the same calculations for all planets and moon in the solar system
7. ignore sun heat and radiation, do the same with the sun
8. other case : how close a satellite be from the sun ? can a satellite orbit around it ? will it susstain the heat, the radiation form the sun ?